Faster Perturbed Stochastic Gradient Methods for Finding Local Minima

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The paper in one minute

Question: Can simple stochastic-gradient perturbations match strong second-order rates without a separate negative-curvature oracle?
Model/Method: LENA (Last stEp shriNkAge) alternates normalized-descent and escape phases, using recursive SPIDER or STORM gradient estimates and controlled step-length shrinkage.
Finding: Under stochastic smoothness and bounded-error assumptions, LENA-SPIDER and LENA-STORM reach an $(\varepsilon, \varepsilon_H)$ approximate second-order point in $\widetilde{O}(\varepsilon^{-3} + \varepsilon_H^{-6})$ stochastic gradients (problem constants hidden).
Why we care: It shows that the difficulty in combining perturbations with variance reduction is controlling estimator error *during escape*, not merely injecting noise.

1. The problem: an inaccurate direction near a saddle

For a stochastic objective $F(x) = \mathbb{E}_\xi f(x;\xi)$ we can sample $\nabla f(x;\xi)$ but cannot cheaply calculate $\nabla F(x)$. A candidate approximate local minimum satisfies

$$\|\nabla F(x)\| \le \varepsilon, \qquad \lambda_{\min}(\nabla^2 F(x)) \ge -\varepsilon_H.$$

A small true gradient may mean a good minimum or a saddle. Perturbed GD separates these cases by adding a random kick at small gradients; the difficulty for stochastic methods is knowing whether the gradient is *actually* small when the estimator is noisy.

Build the gradient estimate first

A SPIDER/SARAH recursion compares a sampled component at consecutive points using *the same* sample:

$$d_t = d_{t-1} + \frac{1}{b}\sum_{i\in B_t}\big[\nabla f_i(x_t) - \nabla f_i(x_{t-1})\big].$$

Its error depends on how far the iterates have moved since the reference estimate, not only on how many steps elapsed. In the paper, a key estimate relates squared gradient error to accumulated squared movements:

$$\|d_t - \nabla F(x_t)\|^2 \lesssim \sum_{j=s_t}^{t-1}\|x_{j+1} - x_j\|^2.$$

The suppressed constants, logs and probability conditions are important for the formal proof (Eq. 4.1).

Worked example: why small steps improve the estimate

Take the illustrative components $f_1(x) = \frac{1}{2}(x-1)^2$, $f_2(x) = \frac{1}{2}(x+1)^2$. The exact gradient of their average is $F'(x) = x$. Start at $x_0 = 2$ with $d_0 = 2$ and move to $x_1 = 1.8$. For either sampled component,

$$f_i'(1.8) - f_i'(2) = -0.2, \qquad d_1 = 2 - 0.2 = 1.8.$$

The recursion reconstructs the exact full gradient here even though the individual gradients differ by 2. In general this exact cancellation does not hold, so restricting cumulative movement matters.

2. LENA: shrink the *last step*, not the objective

The two phases in Algorithm 1

GD phase: When $\|d_t\|$ is large, make a normalized update of controlled length η; this permits progress without letting estimator error grow unpredictably. When the estimated gradient is small, sample a perturbation and begin the escape phase. **Escape phase:** Use the recursive gradient estimator while enforcing a bound on average squared motion. At a trigger involving cumulative displacement, LENA shrinks the *last update* so that total movement remains below its threshold. A sufficiently successful escape produces a noticeable function decrease; otherwise the algorithm returns an approximate second-order stationary candidate.

Why the shrinkage matters mathematically

Imagine three proposed step lengths 0.03, 0.04 and 0.05. Their total squared motion is

$$0.03^2 + 0.04^2 + 0.05^2 = 0.005.$$

If an estimator-accuracy argument allows only 0.003, one can shrink the *last* displacement to $\sqrt{0.003 - 0.03^2 - 0.04^2} \approx 0.0224$. This is a pedagogical analogy to the paper's average-movement safeguard, not its literal parameter rule.

What the theorems establish

Theorems 5.1 and 5.3 analyze LENA with SPIDER and STORM estimators. Their stochastic-gradient complexity is

$$\widetilde{O}\left(\varepsilon^{-3} + \varepsilon_H^{-6}\right),$$

with objective-gap, noise, gradient/Hessian smoothness and confidence factors suppressed. Under $\varepsilon_H = \sqrt{\varepsilon}$, this becomes $\widetilde{O}(\varepsilon^{-3})$. The STORM version admits a single-loop implementation while SPIDER refreshes its estimator periodically.

> **Do not conflate guarantees:** These are stochastic *gradient evaluation* counts, not necessarily wall-clock or iteration counts; the result is approximate second-order stationarity, not global optimality.

Read original Sections 3–6; inspect Algorithm 1, Theorems 5.1 and 5.3, and the proof discussion around Eq. (4.1).

3. Contributions, assumptions, limits, survey role

Contributions. A simpler perturbed stochastic-gradient framework with near-optimal ε^{-3} complexity at the standard $\sqrt{\varepsilon}$ curvature tolerance; explicit control of recursive estimator error by limiting iterate displacement; two implementations, SPIDER and STORM, with different loop structure. **Assumptions.** Every sampled loss is L-smooth with ρ-Lipschitz Hessian (Assumption 3.1); stochastic gradient error is uniformly bounded by σ (Assumption 3.2); and the initial objective gap is finite (Assumption 3.3). Perturbation sizes, batches and stopping thresholds must satisfy the theorem choices. **Limitations.** Assumptions on every sample are strong. The escape guarantee can require carefully chosen parameters. A second-order point need not be globally optimal; online oracle complexities are not directly comparable with finite-sum complexities. The empirical comparisons are task-specific.

Connections to other papers

Jin (2017) introduced explicit perturbations for deterministic GD. **SSRGD (Ref 83)** also perturbs a recursive estimator but with a different analysis and $\widetilde{O}(\varepsilon^{-3.5})$ classical tolerance rate. **SpiderBoost (Ref 80)** focuses primarily on first-order convergence; LENA adds second-order escape. **NEON2 (Ref 77)** detects negative curvature explicitly, whereas LENA avoids an additional curvature-finding module.

Quick reference

Concept	Interpretation
$(\varepsilon, \varepsilon_H)$ point	Gradient $\leq \varepsilon$, negative curvature no worse than $-\varepsilon_H$.
LENA-SPIDER	Periodically refreshed, recursive gradient estimator.
LENA-STORM	Single-loop recursive estimator with momentum-like weights.
Step shrinkage	Caps averaged movement to control stochastic estimator error.
$\widetilde{O}(\varepsilon^{-3} + \varepsilon_H^{-6})$	Stochastic-gradient oracle complexity under paper assumptions.

Papers Master note

Chen–Zhou–Gu's LENA combines variance-reduced stochastic gradients, occasional perturbations and last-step shrinkage; controlling movement keeps gradient estimates reliable during saddle escape. For $\varepsilon_H = \sqrt{\varepsilon}$, it obtains $\widetilde{O}(\varepsilon^{-3})$ oracle complexity under smoothness and bounded-noise assumptions.

Source: official ALT 2022 paper, Assumptions 3.1–3.3, Algorithm 1, Theorems 5.1/5.3; example is pedagogical.